# Archaeomagnetism: A Preliminary Report on Britain

by R. M. COOK and J. C. BELSHÉ

*In this article Mr R. M. Cook, Reader in Classical Archaeology in the University of Cambridge, and Mr J. C. Belshé, Senior Assistant in Research in the Department of Geodesy and Geophysics in the University, discuss the results of the researches into the archaeological significance of remanent magnetism dating which they have been doing during the last five years. Their work continues, and if any reader knows of suitable material they would be grateful if he would get in touch with them through Mr Belshé, Department of Geodesy and Geophysics, Madingley Rise, Madingley Road, Cambridge (Telephone: Cambridge 2103).*

## 1. Introduction

THE existence of permanent magnetism in igneous rocks and fired clays has been known since the middle of the 19th century. Several attempts have been made to use it for geological and archaeological purposes (see Bibliography). The inquiry reported here was begun in 1953 in ignorance that Prof. E. Thellier was already working along similar lines in France.[1] Later we were able to confer with him, learning much from his technical methods, and in return offering him additional material for study. The collaboration has also afforded an opportunity to check the accuracy of our own measurements.

## 2. Theory

Many rocks—that is, according to geological usage, stones and natural clays—contain oxides of iron and are feebly but appreciably magnetic.[2] The character of their magnetism varies according to their nature. If igneous or formed by processes of slow sedimentation in water, a rock may have a natural remanent magnetism which is uniform throughout the layer or deposit and also in the direction of the magnetic field in which it formed. If formed in more disturbed conditions, such as is the case with glacial clays, this natural magnetism may be quite random. Working and moulding material from a uniformly magnetized clay deposit will also tend to make its magnetism random.

When subjected to heat the magnetism of a rock decreases and vanishes at what is called the Curie-point, a temperature usually in the range 500–600° C. If the rock is then cooled below the Curie-point it becomes remagnetized, and the new magnetization has the direction and proportionately the intensity of the magnetic field of its environment. The activities of man have often resulted in clays or stones being fired in this manner.

The effects may be complicated if the material is fired more than once, since a partial magnetization can be acquired even at temperatures below the Curie-point.[3] If the last

[1] See Bibliography, nos. 5, 6, 7.

[2] This ranges from $10^{-2}$ emu/cm$^3$ downwards to intensities of magnetization below any useful level of measurement (approximately $10^{-8}$ emu/cm$^3$). In most of the specimens we have studied the magnetization lies in the range $10^{-2}$—$10^{-6}$ emu/cm$^3$. This is comparable to the magnetization of igneous rocks.

[3] This may reflect the presence of several magnetic components in the material since each will have its characteristic magnetization. It also reflects the balance existing between the magnetic forces favouring alignment of the elementary particles of matter and the kinetic forces disrupting this alignment. Since these kinetic forces are directly proportional to temperature, a small increase in temperature may cause a small loss of magnetic coherence. This effect is, of course, most marked in the temperature range lying near the Curie-point.



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heating reaches the Curie-point, traces of any preceding magnetization vanish. But if the last heating does not reach the Curie-point nor yet the temperature of a previous heating, the rock may possess two (or more) remanent magnetizations and the apparent magnetization will be an intermediate one. Such magnetizations can be detected experimentally by heating a specimen of the rock when it is isolated from any magnetic field. If, for instance, the rock was first heated to 800° C. and later heated again to 400° C., any second magnetization can be destroyed by heating slightly above 400° C. and allowing it to cool in zero field, that is in isolation from the earth's field or other magnetic disturbances. The first magnetization can then be ascertained, and by comparison with the record of the intermediate magnetization the second magnetization can be calculated.

The magnetic field which causes the remanent magnetism of fired rocks is normally that of the earth; this was only to be expected and seems confirmed by results. Other causes are comparatively unimportant and might be deduced if they have occurred in some instances. (1) A strike of lightning may completely remagnetize a specimen. If the direction the electric current might travel is random, then the chances that its resulting magnetization would fall within the possible range of direction of the geomagnetic field are about 1 in 200 and the chances that two distinct structures of the same date should receive a similar magnetization are negligible. (2) Local, natural, magnetic anomalies might distort the earth's field; but those of such magnitude as to be significant for our study should be known from other geophysical investigations and are unlikely to have changed with time. (3) The proximity of iron objects—man-made anomalies—might have a limited effect; but any such objects would hardly be large in size and since their effect decreases with the cube of the distance, samples from different parts of a structure should betray it. Besides, iron was relatively scarce before the modern era. (4) The daily variation in the earth's field, which naturally affected the magnetization of specimens, is usually under 10′, but occasionally as much as 1°. This variation cannot be discounted, though it is usually smaller during the night, when (as we suppose) ancient kilns were usually cooling through their Curie-point. But it sets a useful limit to the accuracy to be attempted.

At any one place the earth's magnetic field varies with time in direction and intensity, so that the remanent magnetization of a specimen should be characteristic of some particular time or times in the earth's history at that place. As an example of this variation in direction of the magnetic vector, FIG. 1 shows the observed change in the annual mean direction of the earth's magnetic field at London from a little before 1600 A.D. to the present day.[4] Since such observations do not go back beyond the 16th century A.D., it is clear that much new information about the earlier history of the earth's magnetic field can be obtained by measuring the direction of the remanent magnetization of suitable rocks, provided that we know the orientation of these rocks when they received their magnetization.

### 3. Aims and Application

Our aims have been twofold. First, we proposed to take samples of fired clay or stone which acquired their remanent magnetization at dates which were known from historical records or archaeological contexts: in this way we hoped to construct for geophyscists a curve of recent variations of the direction of the earth's magnetic field. Secondly, if such a curve could be constructed, we intended to date by their remanent magnetization samples

[4] The direction of the magnetic vector is described here by two angles, those of inclination and declination. The angle of declination, familiar to those who use an ordinary compass, is the horizontal angle between geographic north and the vertical plane that passes through the magnetic vector. The angle of inclination (or dip) is the vertical angle between the vector and the horizontal plane (or tangent) of the earth's surface at the point of measurement.



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which archaeologists and historians could not date by contexts or written records. Archaeological remains that possess a datable remanent magnetism include pots of fired clay, kilns, hearths, and other structures of clay or stone that have been fired whether by accident or design.[5]

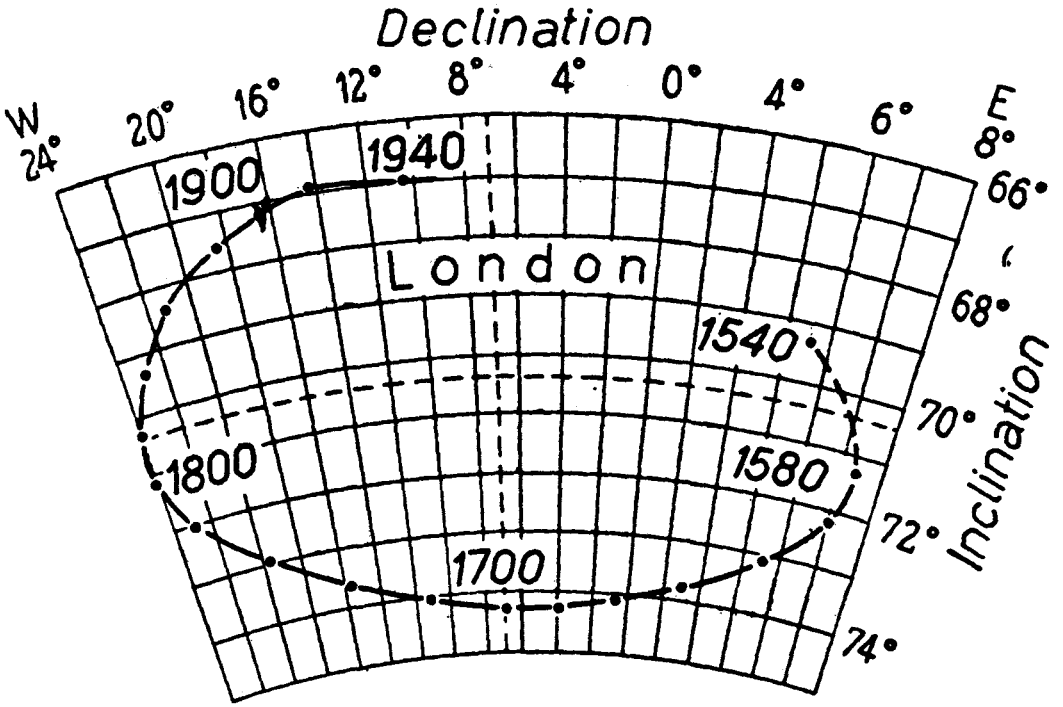



FIG. 1. Observed changes in the annual mean direction of the earth's magnetic field at London since 1580 A.D. The continuation to 1540 A.D. is extrapolated from observations at Rome. (After L. A. Bauer.)

Attempts to use archaeological remains have, of course, been made several times in the last sixty years. But before Thellier's work none was successful. The reason is that the earlier investigators restricted themselves to pots. They conceded that they could not immediately determine from a pot what was its angle of declination (or compass-bearing) at the time when it was fired. But they assumed that the angle of inclination (or dip) was fixed, since pots stood level on their bases when stacked in the kiln. This may be true of glazed ware, where only the unglazed base or spurs had contact with other objects; even so, a pot may balance on its base and yet its axis be several degrees out of the vertical.[6] But most ancient pots were not glazed and, since there was no danger of their fusing together, they could be stacked at any angle in the kiln. Tests that we made on a few random

[5] There are also some dated flows of lava (see Bibliography, no. 2).

[6] In the last two years the Research Laboratory for Archaeology and the History of Art at Oxford has been examining the direction of the remanent magnetism in Chinese porcelain. The results, now published by M. J. Aitken in *Archaeometry*, 1 (1958), 16–20, show a fair consistency.



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specimens of ancient Greek pottery suggest that they were so stacked.[7]

For this reason we decided at the beginning of our inquiry to concentrate on structures which had not been disturbed since they were fired and magnetized. The walls and bottoms of kilns have proved very satisfactory. In firing they have been subjected to a heat of not less than 800° C. Since they are commonly built in pits about 2 ft. deep, their lower parts are often well preserved, and because of their shape and nature it is fairly easy to decide whether those parts have shifted since firing.[8] Their period of use was generally short, so that the remanent magnetization, if not that of the last firing, accumulated over only a few years. Lastly, kilns are fairly numerous in this country, especially of the Roman period, and normally several examples are excavated every year.

## 4. Methods

In the measuring of samples there are two technical problems. First, when a sample has been chosen and before it is removed from the site, its horizontal and vertical bearings must be marked, so that its original orientation can be ascertained when it is measured in the laboratory. Secondly, a suitable instrument is required for making measurements of the remanent magnetization. Both the marking of the bearings and the measuring need to be accurate; in each case the limit we set ourselves was one degree. Suitable techniques had already been developed by students of the natural magnetization of rocks, but during the course of our work we learnt to modify them.

We have used three methods of marking the orientation on samples before removing them. Each method has its own special advantages, but the third is generally the most satisfactory and we now use it whenever we can.

1. If a sample already presents a plane surface, a horizontal line can be scratched or painted on it with the aid of a spirit level. Next, the angle of dip of this surface is determined by an inclinometer along a line at right-angles to the horizontal line. Lastly, the bearing of the horizontal line is taken either by a compass or a theodolite. This method is most useful for some brittle specimens such as tiles.

2. If a sample does not have a suitable flat surface, a small tripod can provide a substitute. This tripod is a plate mounted on three legs, two of which screw with a fine thread. By adjusting the screws the surface of the plate can be made level. A bearing is taken along two fixed points on the plate. The places where the legs touch the sample must be marked and the adjustments of the screws recorded. Then in the laboratory the original orientation can be recovered by adjusting and replacing the tripod in the same position as before.

3. The method we prefer is to cast the sample in a duraluminium mould $4\frac{1}{2}$ in. or 8 in. square and from $2\frac{1}{2}$ in. to 4 in. high. The chosen sample is prepared by cutting round and trimming its sides. The mould is then fitted round the sample and levelled with a spirit level. Next plaster is poured in, enclosing the sides and top of the sample. The top surface of the plaster is carefully made level with the top of the mould, so that it too is horizontal.

[7] The determination of the magnetic vector suggests two limited applications in the study of sherds of wheel-made pottery. Since the wheelmarks are regularly in a horizontal plane—that is, parallel to the plane of the base of the pot—the vertical angle of a sherd may be ascertained. It should then be possible to discover, by comparing the inclination relative to the horizontal plane of the wheelmarks, whether two sherds can or cannot be from the same pot; if they are, the inclination should be the same. Further, if two sherds are from the same pot, their respective declinations should show their relative longitude on the curved surface of the pot. But both these applications are likely to prove difficult in practice.

[8] The cracking and crazing visible in the surface of the walls of many well-preserved ancient kilns probably occurred in the firing of the kiln and so may be, in part at least, earlier than their present magnetization. We judge from observation of the kiln of Roman-British type which F. J. Watson made and fired at Wattisfield in 1956.



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When the plaster has set, an alidade is laid on its surface and a bearing is taken along the line of the alidade and scored on the plaster. After the sample has been removed by cutting away below, the bottom too may be filled in with plaster, so that the sample is completely enclosed in a plaster cube. Specimens prepared by this method can be transported more safely and are more convenient for the magnetometer we now use.

In taking a bearing along the line of the alidade, a theodolite is more reliable than a compass. The theodolite can be used to determine true north by observing the sun angle at the time read by an accurate chronometer or by resection from objects identified on a map. If a compass has to be used, it should be set some distance away from the sample, so that the sample's magnetic moment does not disturb the compass setting. Usually the bearings of several samples can be taken from one station. It is desirable to have at least six samples, symmetrically disposed, from any one structure.

The measurement of a relatively weak magnetization can be done with an astatic magnetometer of the type described by Blackett (Bibliography, no. 11).[9] It consists essentially of two magnets of equal intensity mounted in anti-parallel, and set one a few centimetres vertically above the other. Such a system nullifies the directing effect of the earth's magnetic field acting on a single magnet, since the positive effect of that field on one magnet is balanced by its negative effect on the other This magnet system is hung on a weak torsion fibre which, in the absence of any disturbing forces, determines the rest position of the system. The method of measuring the sample is to bring it, set in some particular orientation, below the lower magnet of the pair. Since the force of the sample's magnetic field decreases by the cube of the distance, the magnetic effect of the sample is much greater on the lower than on the upper magnet. This force causes a deflection of the magnet system against the restoring force of the torsion fibre. From the deflections obtained for several orientations of the sample the magnetic vector best representing the whole sample can be calculated and its direction relative to the orientation plane determined.

At first, to suit the magnetometer we were then using, measurement was done only on small cylinders or discs $1\frac{3}{8}$ in. in diameter, cut from the samples by non-magnetic stone-cutting tools.[10] This process had four dangers—inaccuracy in transferring the marks of the orientation to the cylinder, wobbling in the cutting, the shortness of the orientation marks, and possibly partial remagnetization due to heat applied in drying or generated in hardening. Review of the data obtained by late 1955 confirmed our fears that this method was too imprecise; there was fair accuracy in the measurement of inclination but less for declination. The obvious remedy was to measure the whole sample.

[9] Another type of magnetometer is used by Thellier and the Research Laboratory at Oxford. Here the sample is placed within a set of coils connected to a galvanometer; and, as the sample is rotated, the electric currents which its magnetization generates in the coils are recorded by the galvanometer.

[10] The cutting was done by an electric drill with a diamond corer, and water was used to clear away cuttings and for cooling. This process was satisfactory for hard rocks, but specimens of burnt clay from the walls of kilns are much softer and usually disintegrated in cutting, unless they had been specially prepared. Since it may be useful for other purposes, we describe the method of preparation that we finally adopted. The specimen was allowed to dry, heated to about 100° C. to expel more of its water, and set in an expendable container. Next it was put in a bell-jar, the jar sealed, and the air pumped out by a vacuum pump for about three hours. Then a solution of Marco-resin was introduced into the container till the specimen was covered. After it had soaked for about fifteen minutes the air was let into the bell to assist in forcing the solution to penetrate the specimen. The container was then stood in cold water to reduce the heat given off in the setting of the resin. When it set, the specimen became hard and tough and could be drilled quite safely.

The formula we used was 1 lb. Marco resin SB 280, 44 c.c. Monomer C, 18 grams Catalyst paste H, 14 c.c. Accelerator E.

For our purpose it was convenient to enclose the specimen in Marco resin. If the surface of the object is to be kept free of resin, then the Accelerator should be omitted from the solution and after the interior has been permeated the surface should be wiped clean with Monomer C.



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A visit about this time by Prof. E. Thellier of the University of Paris was very valuable. In spring 1956, on the discovery of the best preserved kiln that had yet been available to us,[11] Prof. Thellier returned to assist in the taking of samples which were to be measured in both laboratories to compare our accuracy. Observation of Thellier's methods convinced us of the merits of casting samples *in situ* and this is now our normal method.

To measure the new cubical blocks a special magnetometer was constructed. The support for the sample is a horizontal table that can be rotated about an axis in line with the axis of the magnet system. In the middle of the support is a square recess which just accommodates the sample. This provides the means for aligning the cube edges with the magnetometer's axes. Large current-carrying coils surround the magnetometer and sample-holder to cancel the earth's magnetic field and avoid induction effects this might cause in the sample. The data obtained, from measurements of all six sides, are plotted on a large stereograph where the vectors can be read to $\frac{1}{2}°$. The data could also be computed mathematically.[12]

PLATE XX A–D shows the remains of a kiln being sampled, a block which has been removed from the mould,[13] and the magnetometer with a sample under it.

## 5. RESULTS

The Table summarizes the results of our measurements of samples collected or received up to August 1957. To make comparisons easier, values for inclination have been adjusted to the latitude of Cambridge in accordance with the data given on Admiralty Chart 5383; we have assumed provisionally that the variations due to latitude only have remained constant (pp. 176–7).

The precision of these results varies. One reason is that the materials of which the structures were made are not all equally satisfactory. Another is that we have not had enough samples from some sites.[14] But the principal reason is that we had to learn by experience; in our earlier work, as is shown by the omission of values for declination, we did not always orientate samples accurately in relation to true north. But later (for example at AGA, AGB, AGC, AGD and AL) the precision has reached the order of 1° (see Bibliography, no. 12). The bar between AF and AGA marks the change from our earlier to our present methods of collection.

Two structures, O and AGE, show notably anomalous results. O had been shifted by the roots of trees, but not enough to account for the lowness of the inclination. More likely explanations are: first, that the samples were bricks from the substructure of the kiln—the only part preserved—and when the kiln was fired the substructure may not have been heated to its Curie-point; secondly, there was above this substructure a grid of iron bars, which may have had a disturbing effect on any remagnetization of the bricks. AGE was

[11] Grimstone End (AGA in the Table).

[12] The result for the sample is obtained by relating this vector to the orientation the sample had *in situ*, and then determining the vector's direction with respect to true north. The result we give for any structure is the mean of all these vector directions. An estimate is made of the probable error from the scatter observed between the results of different samples

[13] At the top on the right is the reference number. Immediately below is the mark of the bearing along the line of the alidade; the arrow-head shows the direction in which the bearing was read. Further down the two parallel lines show magnetic north as determined by a box compass laid on the sample; the direction of north is marked by an arrow-head. See PLATE XX (C).

[14] For example the kilns N and Q*a* were to be preserved, so that we could not remove the specimens we preferred; of the kilns Q*b* and R very little was uncovered; and of B only a small part had survived the bulldozer and exposure.



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a puzzling complex, and the samples from it (which were measured by Thellier, too) gave readings consistent in themselves, but in our opinion very improbable as records of the magnetic field of the earth. Perhaps the magnetization of this kiln was altered by a lightning strike.

Of the other structures AM must be left aside, since it is much the earliest and isolated in time. The remainder, which give generally consonant results, belong to two historical periods, the Roman and the later Medieval. For the Roman period we have enough good samples to allow reasonable certainty about the direction of the magnetic field in England. For the later Medieval period we offer only four structures. FIG. 3 sets out the average readings for declination and inclination of samples from ten Roman and the four Medieval structures. FIG. 2, with measurements of inclination only from thirteen Roman structures, shows the range of variation (or scatter) between different samples from each structure.

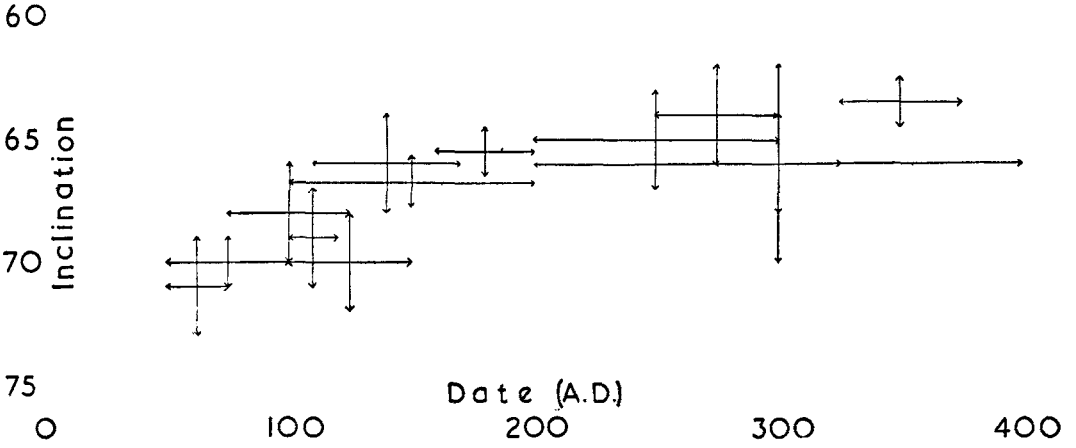



FIG. 2. Scatter in readings of inclination in thirteen samples from Roman kilns. Vertical lines show the scatter of the readings, horizontal the probable latitude in the archaeological dating.

Not all the Roman results are plotted in these diagrams; the selection has been made according to the condition of the structure and the faith we have in our collecting and measuring of the samples. But the inclusion of the omitted results, as the Table shows, does not alter the general trend. During the 1st to the 4th centuries A.D. the direction of the magnetic field shows a steady eastward motion and a shallowing of its dip. In the later Medieval period there appears to be a more rapid motion of increasing inclination as well as an eastward shift in the declination. Both curves conform to the pattern of clockwise rotation recorded by magnetic observatories, and the rate of change agrees reasonably with the observed change of about 2° in twelve years. This rate of change makes it advisable to be cautious in conjectures about the magnetic direction during the long earlier Medieval period, tempting though it is to fill the gap on FIG. 3 with a simple curve.

Our results, both from the Roman and the later Medieval periods, accord with those obtained by Theillier for inclination in northern France and the Rhineland (see Bibliography, nos. 5–6).



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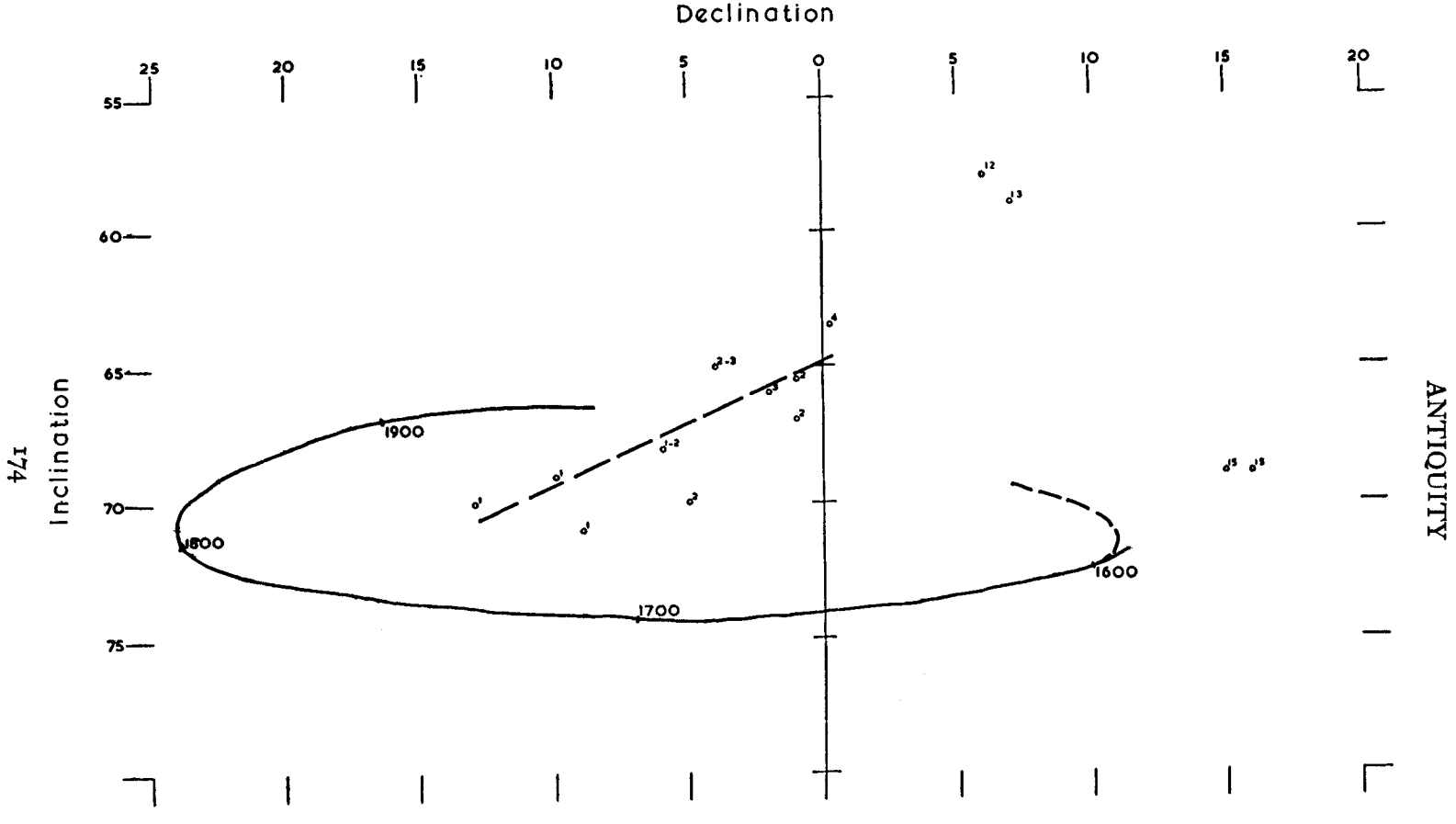



FIG. 3. Changes in the annual mean direction of the earth's magnetic field in England since the 1st century A.D. The continuous black line represents observations at London since 1578 A.D. and the extension in short dashes Bauer's extrapolations from observations at Rome (cf. Fig. 1). The rings show the average readings, corrected for Cambridge, of fourteen of our samples (see Table p. 176-7), and the line of longer dashes the trend for the Roman period.

## 6. Discussion

On FIGS. 1 and 3 the record of direct observations, represented by a solid line, goes back only to the late 16th century A.D. Our points for the 15th century appear as a simple continuation of that line (FIG. 3). This conflicts with the direction suggested by Bauer (FIG. 1).[15] But the relevant part of Bauer's curve (drawn with an intermittent line) is based on extrapolation of observations made in Rome in and after 1540, and he ignored or was ignorant of the observations of Borough and Norman which were made in London in 1576.[16] Since the strong easterly declination about that time may reflect a centre of magnetic disturbance then situated in northern Europe, Bauer's extrapolation is dubious. It is only fair to remark that he himself, unlike some of his followers, did not consider it secure.

It has been known for many years that the earth's magnetic field is characterized by regional centres of magnetic disturbance. Their effects are more clearly exhibited by removing from the observed field that part which would come from the uniformly magnetized sphere most closely fitting the observed field. In 1945 there were seven major centres of disturbance and their regional extent was of the order of a thousand miles.[17] From older observations it is known that such disturbances change their shape and magnitude with time and have slowly drifted westward. The processes causing them are thought to occur in the earth's core and it is their effect on the measurements made by a magnetic observatory which is responsible for the secular change of magnetic components at that observatory.

This regional character of the earth's field means that the results of a study such as ours cannot at once be extrapolated around the world. Even so, the results are more than local in their application and we can hope that, if enough work is done, it should become possible to make valid comparisons of data within regions the size of Europe. For the British Isles our confidence in extrapolating results such as are reported here is still greater.

As has been said, the rate of observed change in the direction of the magnetic components seems to be for Britain about 2° in twelve years. At this rate a measuring accuracy of $\pm$ 1° or even $\pm$ 2° should permit fairly precise chronological computation. If an accurate curve can be constructed, it does not seem over optimistic to expect that suitable material properly sampled and measured should be datable to a quarter of a century.[18]

The accuracy with which the secular change of the earth's magnetic field can be documented is limited principally by the uncertainty of archaeologists in dating their fired structures and, of course, by the supply of such structures. For the Roman period structures that are more precisely dated would be helpful. For the later Medieval period our evidence is gravely insufficient. But even more urgent is the need for material to fill the gap between the 4th and the 13th centuries A.D.

[15] The diagram has often been repeated in text books.

[16] These are marked by the beginning of the solid line on FIG. 3.

[17] The number and extent of such disturbances depend of course on the criteria one chooses to define them. We are generalizing about the foci showing changes in the total magnetic intensity of several tens of gammas per year which have been in evidence for the first half of this century. See figs. 140–143 of Bibliography no. 13.

[18] These estimates apply, of course, to our use of inclination and declination to define the total vector. It would be possible to limit study to one of these and still gain useful knowledge. For example FIG. 2 shows that the variation of inclination during the Roman period might suffice as an aid in dating. Another variable could be added, the intensity of magnetization, which Thellier suggests has recorded a 50 per cent decline during the last 2,000 years (see Bibliography no. 6). Study of intensity alone might be useful for establishing a coarse chronology (measurement to a 10 per cent accuracy would permit dating to about 200 years by Thellier's estimate). Used in conjunction with a study of remanent directions of samples taken *in situ*, the intensity could refine the measurements and decide ambiguities where the curves of direction intersect.



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## TABLE

| Ref. | Location | Type of Structure | Excavators' Date | Condition of Structure | No. of Samples | Declination | Inclination | Comments |
|---|---|---|---|---|---|---|---|---|
| A. | Grimstone End, Pakenham, Suffolk. | Kiln ? | Anglo-Saxon | Probably scattered debris | 4 | Erratic | | Probably not *in situ.* |
| B. | Grimstone End, Pakenham, Suffolk. (No. 19.) | Kiln | 3rd–4th cent. A.D. | Fair | 6 | 2° W (±2) | 66° (±2°) | See Fig. 3. See Fig. 2. |
| C. | Norton, nr Malton, Yorks. (R. H. Hayes and E. Whitley, *Roman Malton and District* Report no. 7, 41–2). | Kiln | Late 3rd–early 4th cent. A.D. | Good | 1 | 10° W (±4°) | 66° (±4°) | Inclination reasonable. See Fig. 2. |
| D. | York (Museum Gardens) | Floor | *c.* A.D. 120 | Fair | 2 | | 60° (±4°) | Inclination shallow. |
| E. | Doncaster (Cantley, kiln 1) . | Kiln | *c.* A.D. 110–170 | Poor | 2 | Broke up | | |
| F. | Doncaster (Cantley, kiln 5) . | Kiln | *c.* A.D. 110–170 | Poor | 1 | Broke up | | |
| G. | Doncaster (Cantley, kiln 4) . | Kiln | *c.* A.D. 110–170 | Poor | 1 | Too weakly magnetized | | |
| H. | Doncaster (Cantley, kiln 3) . | Kiln | *c.* A.D. 110–170 | Perhaps wall had shifted | 1 | 5° W (±4°) | 66° (±2°) | Agrees with date in 2nd cent. A.D. See Fig. 2. |
| J. | Doncaster (Cantley, kiln 2) . | Kiln | *c.* A.D. 110–170 | Fair (kiln had been remodelled once) | 2 | 3° W (±4°) | 59° (±3°) | Inclination very shallow. |
| K. | Rickinghall, Suffolk (Garden House field). | Kiln | 3rd cent. A.D. | Fair | 2 | 5° W (±4°) | 77° (±1°) | Inclination very steep. |
| L. | Overwey, Tilford, Surrey: kiln 1. (A. J. Clark, *Surrey Arch. Coll.*, LI, 29–56.) | Kiln | *c.* A.D. 363–8 | Poor | 3 | Erratic | 78° (±2°) | Inclination very steep. |
| M. | Wattisfield, Suffolk (Mill House Farm). | Kiln | 15th cent. A.D. | Fair | 2 | 16° E (±3°) | 69° (±1°) | See Fig. 3. |
| N. | Wattisfield, Suffolk (Foxledge Common: this kiln was later removed to the Wattisfield Potteries). | Kiln | 2nd–3rd cent. A.D. | Good | 2 | | 76° (±4°) | See n. 14. |
| O. | Lowestoft (112 Yarmouth Road: probably Hewling Luson's first experimental kiln). | Kiln | *c.* A.D. 1756 | Doubtful | 3 | | 49° | See p. 172. |
| P. | Chew Park, Somerset: | | | | | | | |
| | (*a*) Roman lime-kiln . . | Lime-kiln | Probably later 3rd cent. A.D. | Fair | 3 | | 64° (±2°) | See Fig. 2. |
| | (*b*) Medieval lime-kiln no. 2 . | Lime-kiln | 12th cent. A.D. | Fair | 1 | 6° E | 58° | See Fig. 3. |
| | (*c*) Medieval lime-kiln no. 3 . | Lime-kiln | 12th cent. A.D. | Fair | 1 | Too weakly magnetized | | |
| | (*d*) Oven no. 1 . . . | Flue | | Fair | 1 | Too weakly magnetized | | |
| Q. | Sneyd Green, Hanley: (*a*) . | Kiln | *c.* A.D. 1300 | Fair | 2 | 7° E | 59° (±2°) | See Fig. 3. |
| | (*b*) . | Kiln | *c.* A.D. 1300 ? | Doubtful | 1 | Erratic | | |
| R. | Botesdale, Suffolk (prefab. site) . | Kiln | Late 2nd–3rd cent. A.D. | Only edge available | 2 | | 57° (±4°) | Inclination very shallow. |
| S. | Rickinghall, Suffolk (Kiln Farm —see A1). | Kiln | 3rd cent. A.D. | Fair | 2 | 4° W (±3°) | 65° (±2°) | See Fig. 3. See Fig. 2. |
| T. | Totternhoe, nr Dunstable . | Hypocaust flue | | Fair | 3 | Erratic | | |
| U. | High Wycombe, Bucks: (*a*) . | Flue | A.D. 150–200 | | 1 | Too weakly magnetized | | |
| | (*b*) . | Hearth | A.D. 150–200 | | 1 | Too weakly magnetized | | |
| V. | Wattisfield, Suffolk (Hall Farm) | Kiln | Perhaps 3rd cent. A.D. | Poor | 3 | | 69° (±1°) | Inclination suggests 1st–2nd cent. A.D. |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| W. | Kirkstall Abbey, Leeds (kitchen): (*a*) | Hearth | Late 12th–15th cent. A.D. | Poor | 2 | Too weakly magnetized | | | |
| | (*b*) | Hearth | 15th–early 16th cent. A.D. | Poor | 1 | 15° E | 69° (±2°) | See Fig. 3. | |
| X. | Benetfinks, London | Hearth | Later 1st–2nd cent. A.D. | | 1 | 6° W | 68° (±2°) | See Fig. 3. | See Fig. 2. |
| Y. | High Cross, Leicestershire: (*a*) Trench 50 | Burnt area | | Doubtful | 1 | Too weakly magnetized | | | |
| | (*b*) Trench 52 | Burnt area | | Doubtful | 1 | Too weakly magnetized | | | |
| | (*c*) Trench 31 | Flue | | Poor | 2 | Erratic | | | |
| Z. | Woodhouse Hill, Studland, Dorset: (*a*) | Hearth | Probably 4th cent. A.D. | | 1 | Broke up | | | |
| | (*b*) | Hearth | Probably 4th cent. A.D. | | 1 | Too weakly magnetized | | | |
| | (*c*) | Hearth | Probably *c.* A.D. 110 | | 1 | Broke up | | | |
| | (*d*) | Hearth | Probably *c.* A.D. 50 | | 1 | Erratic | | | |
| AA. | Wattisfield, Suffolk (Grundle Farm: higher level—see AGE). | Kiln | Perhaps 13th–17th cent. A.D. | Poor and perhaps disturbed | 3 | 9° E | 57° | Suggests 13th cent. A.D. | |
| AB. | Dounton, nr Salisbury (villa) | Hypocaust flue | A.D. 300–320 | | 1 | | 85° | Inclination too steep. | |
| AC. | Thurgarton, Notts (Castle Hill). (*Journal of Medieval Arch.*, I). | Crucible ? | Later medieval | Good | 1 | Too weakly magnetized | | | |
| AD. | Caer-Mynydd, nr Rhiwlas, Caernarvonshire. | Hearth | | | 1 | Broke up | | | |
| AF. | Grimstone End, Pakenham, Sflk. | Kiln | *c.* A.D. 160–200 | Good | 1 | 2° W | 64° | See Fig. 3. | See Fig. 2. |
| AGA. | | | | | 13 | 1° W (±2°) | 65½° (±1°) | | |
| AGB. | Canterbury II | Tile-kiln | *c.* A.D. 350 | Good | 12 | ½° E (±2°) | 63½° (±1°) | See Fig. 3. | See Fig. 2. |
| AGC. | Canterbury III | Kiln | A.D. 100–120 | Good | 4 | 10° W (±3°) | 69° (±2°) | See Fig. 3. | See Fig. 2. |
| AGD. | West Stow, Suffolk | Kiln | 2nd cent. A.D. | Fair | 9 | 1¼° W (±2°) | 66¾° (±1°) | See Fig. 3. | See Fig. 2. |
| AGE. | Wattisfield, Suffolk (Grundle Farm: lower level—see AA). | Kiln | Perhaps 13th–17th cent. A.D. | Poor and perhaps disturbed | 4 | 84° E (±1°) | 43° (±2°) | See p. 172. | |
| AH. | Upper Sheringham, Norfolk | Kiln | A.D. 100–150 | Good | 5 | 5° W (±3°) | 70° (±2°) | See Fig. 3. | See Fig. 2. |
| AI. | Rickinghall, Suffolk (Kiln Farm—see S). | Kiln | | Poor | 7 | 0° (±5°) (Weakly magnetized and erratic) | 60° (±2°) | | |
| AJ. | Rickinghall, Suffolk | Kiln | | Poor | 4 | Too weakly magnetized | | | |
| AK. | Canterbury IV | Kiln | Later 1st cent. A.D. | Fair | 10 | 13° W (±3°) | 70° (±1°) | See Fig. 3. | See Fig. 2. |
| AL. | Stoke-on-Trent (Trent Vale) | Kiln | A.D. 50–75 | Good | 6 | 9° W (±3°) | 71° (±2°) | See Fig. 3. | See Fig. 2. |
| AM. | Salmonby, Lincs (Holbeck Manor Farm) | Burnt area | Bronze Age | Poor | 2 | Erratic | 70° | See p. 173 | |

NOTES. U, X, Z, AB, and AD were collected by the excavators and sent to us.

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Our experience suggests that satisfactorily magnetized samples may be obtained from a large proportion of fired structures. Most of our work has been done on the walls of kilns. These were the structures most readily available to us and they have proved also the most convenient. But hearths and the remains of burnt buildings are useful too and can often be dated more exactly.

A study of this kind depends on the help of field archaeologists, and we hope that we shall continue to profit by it. Some, but not all, of our debts are noted at the foot of the Table. Prof. E. Thellier has been most generous with advice and aid. We owe much, too, to Dr K. M. Creer, who till 1954 was the principal collaborator on the geophysical side and was active in the collection and preparation of samples. Prof. S. K. Runcorn gave helpful counsel in the early stages. Mrs K. Cook has throughout taken part in collecting. Mr S. C. Collard has contributed in various ways, especially in the preparing of specimens; so too has Mr A. Reeves. The Managers of the Dame Mary Grainger Sandys Fund and of the Craven Fund and the Department of Scientific and Industrial Research have contributed towards the cost of the work. For all this and other assistance we express our warm gratitude.

*P.S.* According to their excavators AK is late 1st–2nd cent. A.D., and AF/AGA *c.* A.D. 200–230. These small corrections, received during printing, could not be used in the Figures or Table.

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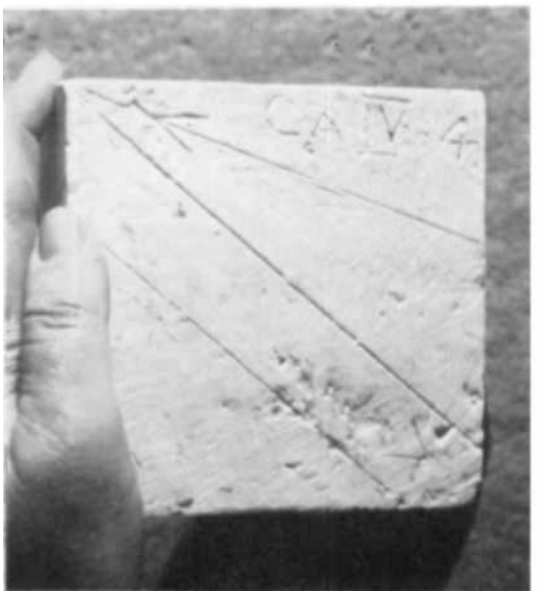
PLATE XX



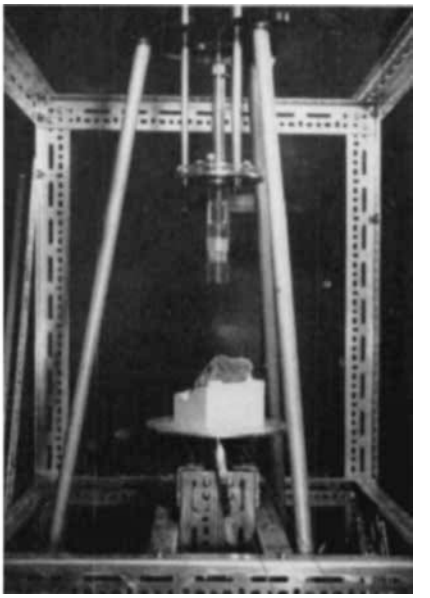
ARCHAEOMAGNETISM. (*a*) Dunnideer, vitrified fort. The samples have been cut round and enclosed in moulds.



(*b*) Filling the same moulds with plaster.



(*c*) Sample detached and freed from mould. *For full description see footnote* 13, *p.* 172.



(*d*) Inner structure of magnetometer with sample on table.

[*See p.* 171

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